2)

Momentum :

17B.2) K) $\sum_{\alpha=1}^{N} j_\alpha = 0$

$\rightarrow \sum_{\alpha=1}^{N} j_\alpha = \sum_{\alpha}^{N} c_\alpha (v_\alpha - v) = 0$

$\hookrightarrow \sum_{\alpha}^{N} c_\alpha v_\alpha - v \sum_{\alpha}^{N} c_\alpha = 0 \rightarrow \rho v - v \rho = 0$

O) $\sum_{\alpha=1} J_\alpha^* = 0 = \sum_{\alpha}^{N} c_\alpha (v_\alpha - v^*) = \sum_{\alpha}^{N} c_\alpha v_\alpha - v^* \sum_{\alpha}^{N} c_\alpha$

$\hookrightarrow cv^* - v^* c = 0$

T) $j_\alpha = n_\alpha - \omega_\alpha \sum_{\beta=1}^{N} n_\beta$

$\hookrightarrow j_\alpha = \rho_\alpha (v_\alpha - v) = \underbrace{\rho_\alpha v_\alpha}_{N_\alpha} - \underbrace{\rho_\alpha v}_{\omega_\alpha \rho \cdot v}$ & $\omega_\alpha = \frac{\rho_\alpha}{\rho}$

$\hookrightarrow j_\alpha = N_\alpha - \omega_\alpha \sum_{\beta}^{N} N_\beta$

X) $J_\alpha^* = N_\alpha - x_\alpha \sum_{\beta=1}^{N} N_\beta$

$\hookrightarrow J_\alpha^* = c_\alpha (v_\alpha - v^*) \rightarrow J_\alpha^* = \underbrace{c_\alpha v_\alpha}_{N_\alpha} - c_\alpha v^*$

$x_\alpha = \frac{c_\alpha}{c}$ & $cv^* = \sum_{\beta=1}^{N} N_\beta$

$\hookrightarrow J_\alpha^* = N_\alpha - x_\alpha \sum_{\beta=1}^{N} N_\beta$

17B.3)

$$\frac{j_A}{\rho w_A w_B} = \frac{J_A^*}{c x_A x_B} \rightarrow \qquad c x_A = c_A \quad \& \quad \rho_A = \rho w_A$$

$$j_A = \rho_A (v_A - v) \quad \& \quad J_A^* = c_A (v_A - v^*)$$

$$\& \quad v = w_A v_A - w_B v_B \qquad v^* = c_A v_A - c_B v_B$$

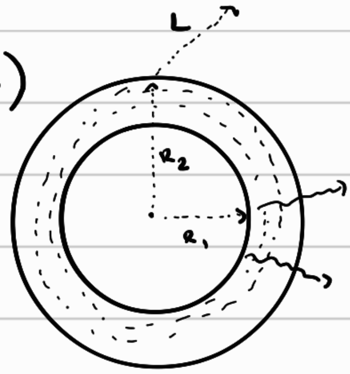
$$\hookrightarrow \quad j_A = \rho w_A (v_A - w_A v_A + w_B v_B) \quad \& \quad J_A^* = c x_A (v_A - c_A v_A + c_B v_B)$$

$$\hookrightarrow \quad j_A = \rho w_A (v_A \overbrace{(1 - w_A)}^{w_B} + w_B v_B) \qquad J_A^* = c x_A (v_A \overbrace{(1 - c_A)}^{c_B} + c_B v_B)$$

$$\hookrightarrow \quad j_A = \rho w_A w_B (v_A + v_B) \quad \& \quad J_A^* = c x_A x_B (v_A + v_B)$$

$$\hookrightarrow \quad \boxed{v_A + v_B = v_A + v_B}$$

18B.8)



$$C_A = C_A(r)$$

$$\langle N_A| = \cancel{C_A \langle v^*|} + \langle J_A^*|$$

↳ no velocity :)

↳ $\langle N_A| = \langle J_A^*| \rightarrow \langle J_A^*| = -c D_A \langle \nabla x_A|$

↳ $\langle J_A^*| = -D_{AB} \langle \nabla C_A)$

↳ $\langle J_A^*| = -D_{AB} \dfrac{dC_A}{dr}$

Shell volume: $L \cdot 2\pi \cdot r \cdot \Delta r$

↳ $[L \cdot 2\pi r \, N_{Ar}]_r - [L \, 2\pi r \, N_{Ar}]_{r+\Delta r} = 0$

↳ divide by shell vol → $\lim \Delta r \rightarrow \infty$

↳ $\dfrac{d}{dr}(r N_{Ar}) = 0 \rightarrow D_{AB} \dfrac{d}{dr}\left(r \cdot \dfrac{dC_A}{dr}\right) = 0$

↳ $r \cdot \dfrac{dC_A}{dr} = C_1 \rightarrow \dfrac{dC_A}{dr} = \dfrac{C_1}{r} \rightarrow C_A = C_1 \ln(r) + C_2$

$C_A(R_1) = C_{A1} \rightarrow C_{A1} = C_1 \ln(R_1) + C_2$

$C_A(R_2) = C_{A2} \rightarrow C_{A2} = C_1 \ln(R_2) + C_2 \rightarrow C_2 = C_{A2} - C_1 \ln(R_2)$

↳ $C_{A1} = C_1 \ln(R_1) + C_{A2} - C_1 \ln(R_2)$

$C_{A1} - C_{A2} = C_1(\ln(R_1) - \ln(R_2)) \rightarrow C_{A1} - C_{A2} = C_1 \ln \dfrac{R_1}{R_2}$

↳ $C_1 = \dfrac{C_{A1} - C_{A2}}{\ln(R_1/R_2)}$ & $C_2 = C_{A2} - \dfrac{C_{A1} - C_{A2}}{\ln(R_1/R_2)} \ln(R_2)$

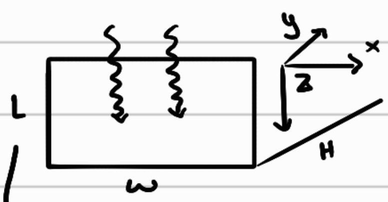
↳ $N_A = -D_{AB} \dfrac{dC_A}{dr} = -D_{AB} \cdot \dfrac{C_{A1} - C_{A2}}{\ln(R_1/R_2)} \cdot \dfrac{1}{r} = N_{A,r}$

↳ $N_A = \dfrac{D_{AB}(C_{A1} - C_{A2})}{\ln(R_2/R_1)} \cdot \dfrac{1}{r}$

↳ $W_A = N_A \cdot A = N_A \cdot 2\pi r \cdot L$

↳ $$\boxed{W_A = \dfrac{D_{AB}(C_{A1} - C_{A2})}{\ln(R_2/R_1)} \cdot 2\pi L}$$

18B.5)



$r_A = -k_2''' C_A^2$

air = insoluble!

can be taken as ∞

a) $N_A = J_A^* = -D_{AB} \dfrac{dC_A}{dz}$

$$\left[W \cdot H \cdot N_{Az}\right]_z - \left[WH\, N_{Az}\right]_{z+\Delta z} + k_2''' C_A^2 \cdot \Delta z\, WH = 0$$

↳ ÷ shell vol → lim $\Delta z \to 0$

$$↳ -\frac{dN_{Az}}{dz} = k_2''' C_A^2 \rightarrow \frac{d^2C_A}{dz^2} = \frac{k_2''' C_A^2}{D_{AB}} \rightarrow$$

$$\frac{dC_A}{dz} = P \rightarrow \frac{dP}{dz} = \frac{dP}{dC_A} \overbrace{\frac{\partial C_A}{\partial z}}^{P} \rightarrow \frac{dP}{dz} = P\frac{dP}{dC_A}$$

$$↳ P\frac{dP}{dC_A} = \frac{k_2''' C_A^2}{D_{AB}} \rightarrow P\,dP = \frac{k_2'''}{D_{AB}} C_A^2\, dC_A \rightarrow \frac{1}{2}P^2 = \frac{k_2'''}{3D_{AB}} C_A^3 + C_1$$

$$↳ \frac{1}{2}\left(\frac{dC_A}{dz}\right)^2 = \frac{k_2'''}{3D_{AB}} C_A^3 + C_1$$

@ $z = L \rightarrow \dfrac{dC_A}{dz} = 0$ & $C_A = 0 \rightarrow 0 = C_1$

$$↳ \frac{1}{2}\left(\frac{dC_A}{dz}\right)^2 = \frac{k_2'''}{3D_{AB}} C_A^3 \rightarrow \frac{dC_A}{dz} = -\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}} \cdot C_A^{\frac{3}{2}}$$

$$↳ C_A^{-\frac{3}{2}}\, dC_A = -\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}} \cdot dz \rightarrow 2C_A^{-\frac{1}{2}} = -\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}} \cdot z + C_2$$

$$2C_{A,0}^{-1/2} = -\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}} \cdot 0 + C_2 \rightarrow C_2 = 2C_{A,0}^{-1/2} + \cancel{\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}}\, 0} \rightarrow C_2 = 2\frac{1}{C_{A0}^{1/2}}$$

$$↳ 2\frac{1}{\sqrt{C_A(z)}} = -\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}}\, z + 2\frac{1}{\sqrt{C_{A,0}}} \rightarrow \boxed{\frac{C_{A,0}}{C_A} = \left[1 + \sqrt{\frac{k_2''' C_{A,0}}{6 D_{AB}}}\, z\right]^2}$$

b) $N_{Az}\big|_{z=0} = -D_{AB}\cdot \frac{\partial c_A}{\partial z}\big|_{z=0}$

& $\frac{dc_A}{dz} = -\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}}\cdot c_A^{\frac{3}{2}}$

↳ $N_{Az}\big|_{z=0} = -D_{AB}\left(-\sqrt{\frac{2}{3}\frac{k_2'''}{D_{AB}}}\; c_A^{3/2}\right)$

↳ $\boxed{N_{Az}\big|_{z=0} = \sqrt{\frac{2}{3}D_{AB}k_2'''}\; c_A^{3/2}}$

c) $[WH\,N_{Az}]_z - [WH\,N_{Az}]_{z+\Delta z} - f(c_A) = 0$

$-\frac{dN_{Az}}{dz} = f(c_A) \rightarrow \frac{dN_{Az}}{dz} = -f(c_A) \rightarrow -D_{AB}\frac{d^2c_A}{dz^2} = -f(c_A)$

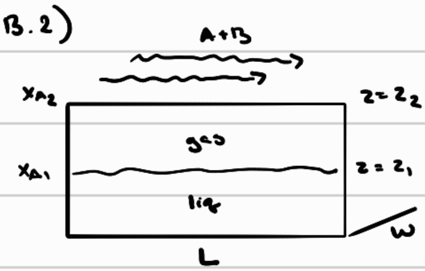
$\frac{dc_A}{dz} = p \rightarrow p\frac{dp}{\partial c_A} = \frac{f(c_A)}{D_{AB}} \rightarrow p\,dp = \frac{f(c_A)}{D_{AB}}dc_A \rightarrow \frac{1}{2}p^2 = \int \frac{f(c_A)}{D_{AB}}dc_A + C$

$\frac{dc_A}{dz} = 0$ & $c_A = 0$ @ $z = L$

↳ $\left(\frac{dc_A}{dz}\right)^2 = 2\int \frac{f(c_A)}{D_{AB}}dc_A \rightarrow \frac{dc_A}{dz} = -\sqrt{2\int \frac{f(c_A)}{D_{AB}}dc_A}$

↳ $N_{Az}\big|_{z=0} = -D_{AB}\left(-\sqrt{2\int\frac{f(c_A)}{D_{AB}}dc_A}\right) \rightarrow \boxed{N_{Az}\big|_{z=0} = \sqrt{2D_{AB}\int f(c_A)\,dc_A}}$

18B.2)



$N_{Bz} = 0$

$N_{Az} = -cD_{AB}\frac{dx_A}{dz} + \underbrace{\cancel{x_A(N_{Az}+N_{Bz})}}_{\text{neglect}}$

↳ $N_{Az} = -cD_{AB}\frac{dx_A}{dz}$

$[WL\cdot N_{Az}]_z - [WL\,N_{Az}]_{z+\Delta z} = 0 \rightarrow -\frac{dN_{Az}}{dz} = 0 \rightarrow \frac{dN_{Az}}{dz} = 0$

↳ $-cD_{AB}\frac{d^2x_A}{dz^2} = 0 \rightarrow \frac{d^2x_A}{dz^2} = 0 \rightarrow \frac{dx_A}{dz} = C_1 \rightarrow x_A = C_1z + C_2$

$x_A\big|_{z=z_1} = x_{A1} \rightarrow x_{A1} = C_1z_1 + C_2$ $\qquad x_{A2} = C_1z_2 + C_2$

↳ $C_2 = x_{A2} - C_1z_2 \rightarrow x_{A1} = C_1z_1 + x_{A2} - C_1z_2$

$$\hookrightarrow \quad x_{A_1} - x_{A_2} = c_1 (z_1 - z_2) \rightarrow c_1 = \frac{x_{A_1} - x_{A_2}}{z_1 - z_2}$$

don't need

$$\hookrightarrow \quad \cancel{c_2 = x_{A_2} - \frac{x_{A_1} - x_{A_2}}{z_1 - z_2} z_2}$$

$$\hookrightarrow \quad N_{Az} = -D_{AB} \cdot c \cdot \frac{dx_A}{dz} \rightarrow N_{Az} = -c D_{AB} \cdot c_1 = \boxed{N_{Az} = c D_{AB} \cdot \frac{x_{A_1} - x_{A_2}}{z_2 - z_1}}$$

b)

$$18.2-14) \quad N_{Az}\Big|_{z=z_1} = -\frac{c D_{AB}}{1 - x_{A_1}} \frac{dx_A}{dz}\Big|_{z=z_1} = \frac{c D_{AB}}{x_{B_1}} \frac{dx_B}{dz}\Big|_{z=z_1} = \frac{c D_{AB}}{z_2 - z_1} \ln\left(\frac{x_{B2}}{x_{B1}}\right)$$

$\hookrightarrow$ since we assume x_{A_1} & $x_{A_2} \lll 1$

$$\hookrightarrow \quad N_{Az}\Big|_{z=z_1} = \frac{c D_{AB}}{z_2 - z_1} \ln\left(\frac{1 - x_{A_2}}{1 - x_{A_1}}\right)$$ & for small $x \rightarrow \ln(1-x) = -x$

$$\hookrightarrow \quad N_{Az}\Big|_{z=z_1} = \frac{c D_{AB}}{z_2 - z_1} (-x_{A_2} + x_{A_1}) \rightarrow \boxed{N_{Az}\Big|_{z=z_1} = \frac{c D_{AB}}{z_2 - z_1} (x_{A_1} - x_{A_2})}$$

c) $D_{AB,\log} = 0.0636 \ cm^2/s$

$z_2 - z_1 = 17.1 \ cm$ $\quad N_A = 7.26E-9$

$\hookrightarrow$

$$\hookrightarrow \quad D_{AB} = \frac{N_{Az} \cdot (z_2 - z_1)}{c (x_{A_1} - x_{A_2})} \rightarrow D_{AB} = \frac{N_{Az} (z_2 - z_1) RT}{P \left(\frac{P_{A_1}}{P} - \frac{P_{A_2}}{P}\right)}$$

& $P_B = P - P_A$

$\hookrightarrow P_A = P - P_B$

$$\hookrightarrow \quad D_{AB} = \frac{N_{Az} (z_2 - z_1) RT}{(\cancel{P} - P_{B_1} - \cancel{P} + P_{B_2})} \rightarrow D_{AB} = \frac{N_{Az} (z_2 - z_1) RT}{P_{B_2} - P_{B_1}}$$

$$D_{AB} = \frac{7.29E-9 \cdot 17.1 \cdot 82.06 \cdot 273}{(755 - 722)/760} = 0.06405$$

$$\hookrightarrow \text{error}: \left|\frac{0.6405 - 0.0636}{0.0636}\right| \cdot 100 = \boxed{0.708\%}$$